



The **Testing Development Life Cycle (TDLC)**, more commonly referred to as the **Software Testing Life Cycle (STLC)**, is a structured sequence of phases that ensures software is thoroughly tested before release.

It runs parallel to the **Software Development Life Cycle (SDLC)** and provides a repeatable framework for **planning, designing, executing, and closing testing** activities.



Testing Development Life Cycle (TDLC)

TDLC/STLC is the **process of validating software quality** through systematic phases. Each phase has **entry criteria, activities, exit criteria, and deliverables**, ensuring testing is deliberate and measurable rather than ad-hoc.



Phases of TDLC/STLC

1. Requirement Analysis

- Review functional and non-functional requirements.
- Identify testable features and clarify ambiguities.

Deliverable: Requirement Traceability Matrix (RTM).

2. Test Planning

- Define scope, objectives, resources, schedule, and risk mitigation.

Deliverable: Test Plan document.

3. Test Case Design & Development

- Create test cases, test scripts, and prepare test data.

Deliverable: Test Case Repository.

4. Test Environment Setup

- Configure hardware/software to simulate production conditions.

Deliverable: Stable test environment.

5. Test Execution

- Run test cases, log defects, validate outcomes.

Deliverable: Defect reports, execution logs.

6. Test Cycle Closure

- Evaluate test coverage, defect density, lessons learned.

Deliverable: Test Summary Report.



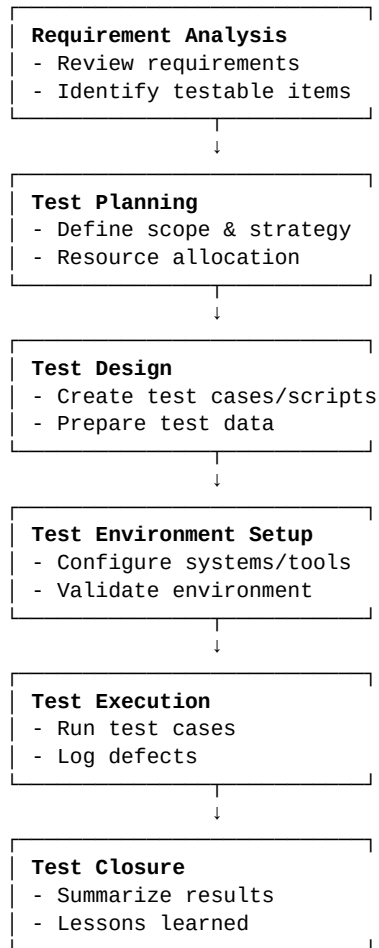
Quick Comparison Table

<u>Phase</u>	<u>Focus</u>	<u>Deliverable</u>
Requirement Analysis	Identify testable features	RTM
Test Planning	Define scope & strategy	Test Plan
Test Case Design	Create cases/scripts	Test Repository
Environment Setup	Configure systems	Test Environment
Execution	Run tests & log defects	Defect Reports
Closure	Evaluate & improve	Test Summary Report

- TDLC/STLC ensures **early defect detection**, reducing cost and risk.
- It aligns testing with **business requirements** and **development phases**.
- In Agile/DevOps, these phases are **iterative and continuous** within each sprint or CI/CD pipeline.



The **Testing Development Life Cycle (TDLC)** showing the six phases in sequence.



Phase Overview

- **Requirement Analysis** → Identify testable features.
- **Test Planning** → Define scope, resources, risks.
- **Test Design** → Build cases, scripts, data.
- **Environment Setup** → Prepare stable test systems.
- **Execution** → Run tests, log defects.
- **Closure** → Evaluate coverage, document lessons.

✨ This flow ensures testing is **systematic and measurable**, reducing risk and improving software quality. Each phase has **entry/exit criteria** so teams know when to move forward.